HW 2

Solutions

Modeling Democracy
Duchin, Winter 2026

Problem 1. Suppose three districts (districts B, P, and R) have populations $M_B = 54, M_P = 243, M_R = 703$. Apportion the seats by Hamilton's method with $m = 10$ and $m = 11$ seats, with no constitutional requirement that everybody gets a seat on the council. Also consider $M'_B = 56, M'_P = 255, M'_R = 789$ at the same council sizes of 10 and 11. Which combinations of these conditions produce apportionment paradoxes?

Solution 1. *For $(M_B, M_P, M_R) = (54, 243, 703)$ and $m = 10$, the quotas are $(0.54, 2.43, 7.03)$. The initial allocation is $(0, 2, 7)$ and Hamilton gives district B one more seat resulting in final allocation $(1, 2, 7)$. For $m = 11$, the quotas are approximately $(0.594, 2.673, 7.733)$ so Hamilton awards P and R one more seat each resulting in final allocation $(0, 3, 8)$. We see the **Alabama paradox** since district B lost a seat when the number of total seats increased.*

*For $(M_B, M_P, M_R) = (56, 255, 789)$, Hamilton results in final allocations of $(1, 2, 7)$ for $m = 10$ and $(1, 2, 8)$ for $m = 11$. We see the **population paradox** for $m = 11$ since the population of P increased by around 4.9%, the population of B increased by around 3.7%, but B stole a seat from P.*

Problem 2. The Balinski–Young impossibility result applies to deterministic apportionment methods. If we are allowed to use randomness, Grimmett showed in 2004 that we can circumvent Balinski–Young. In this problem we construct a random apportionment method that always satisfies the quota rule and achieves perfect proportionality in expectation.

Give each state its lower quota, and then (randomly) award some states one more seat, as follows: Let the fractional parts of the quota for the states be $\delta_1, \ldots, \delta_n \in [0, 1)$, and let k denote their sum, which is the number of seats not yet awarded. We line up n intervals on the number line without gaps, where the ith interval has length δ_i. We then shift all intervals to the right by a random amount u, drawn uniformly from $[0, 1)$. This now produces subintervals that cover $[u, u + k]$. Award an extra seat to exactly those states whose interval contains an integer.

For a state with $q + \delta$ seats, what is the probability that this process will round them up or down? Conclude that the method works as advertised.

Solution 2. *We check that the Grimmett randomized apportionment method satisfies the following properties:*

- ***Perfect proportionality:*** *A state with quota $q + \delta$ is first awarded its lower quota q and the residue is an interval I of length δ. Since $u \in [0, 1)$ is drawn uniformly, the probability that $I + u$ contains an integer is δ.*

  To visualize this, imagine translating interval I back to start at the origin, i.e., translating to the interval $[0, \delta]$. After this translation, one integer falls in the range $[0, 1)$ but it's uniformly random because of the shift by u. Clearly a uniformly random value between 0 and 1 falls in $[0, \delta]$ with probability δ.

  The expected number of additional seats awarded is thus δ.

- ***Quota rule:*** *A state with quota $q+\delta$ is either awarded q seats (if no integer fell in its range) or $q+1$ seats (if one integer fell in its range). Since $\delta < 1$, an interval of length δ can't contain two integers.*

- ***Total number of seats:*** *The sum of the residues is $k \in \mathbb{Z}$. (It's an integer because it adds up to the number of seats not yet awarded.) The shifted subintervals combine to form an interval $[u, u+k)$ of length k. This must contains exactly k integers because of its length, so the number of additional seats awarded is always k.*

Problem 3. We will write $\mathbf{n} = (n_1, \ldots, n_r)$ for the populations of r towns; $\mathbf{w}$ is the vector of voting weights; T is a *strict* threshold of success for a vote (so a winning coalition must have weight $> T$); and $\boldsymbol{\beta}$ is the vector of Banzhaf powers. Assume all vectors are normalized to sum to 1. Write f for the Banzhaf objective $|\mathbf{n} - \boldsymbol{\beta}|_1$ to be minimized.

(a) Prove that if $n_i \geq n_j$, then there is a weight vector at optimum discrepancy with $w_i \geq w_j$.

(b) By contrast, show that if $\mathbf{n}$ has two equal entries $n_i = n_j$, there may be NO optimal solution with $w_i = w_j$. (For example, there exist four-player games with this symmetry-breaking property.)

(c) BONUS: can a four-player game have the symmetry-breaking property at $T = 1/2$? Can any three-player game have the symmetry-breaking property?

Note: for the Banzhaf problem, you should not need to write code to assist you, but you are welcome to do that for the purpose of testing examples. With or without the use of code, a complete solution must include a proof of correctness.

Solution 3. *This was meant to be a hard problem, and kind of open-ended! Here is Alec's quite nice solution. Please ask one of us if you want this spelled out more verbosely.*

(a) Assume for contradiction that all weight vectors at optimum discrepancy satisfy $w_i < w_j$.

PROPOSITION: If $w_i \leq w_j$ then $\beta_i \leq \beta_j$.

Proof. It suffices to construct an injective map f from the set of winning coalitions for which i is a swing vote to the set of winning coalitions for which j is a swing vote. Let S be a winning coalition containing i. Define

$$f(S) \begin{cases} S & \text{if } j \in S \\ S \setminus \{i\} \cup \{j\} & \text{if } j \notin S. \end{cases}$$

Since $w_i \leq w_j$, in both cases $f(S)$ is a winning coalition and j is a swing vote. □

By the proposition, we have $\beta_i \leq \beta_j$. Consider swapping the weights w_i and w_j so that the new weight of j is $w'_j = w_i$ and the new weight of i is $w'_i = w_j$. By symmetry, this swaps β_i and β_j so that $\beta'_i \geq \beta'_j$. Since we assumed $n_i \geq n_j$, we have $\|\mathbf{n} - \boldsymbol{\beta}'\|_1 \leq \|\mathbf{n} - \boldsymbol{\beta}\|_1$ so $\boldsymbol{w}'$ with $w'_i > w'_j$ must also be optimal, contradiction.

(b) Label the towns A, B, C, D in order of weakly decreasing weights. The set of winning coalitions is upward-closed and hence we can enumerate over all possible sets of winning coalitions by enumerating over all possible sets of minimal winning coalitions to compute the possible Banzhaf powers:

Consider normalized populations $\mathbf{n} = \left(\frac{7}{17}, \frac{7}{17}, \frac{2}{17}, \frac{1}{17}\right)$ with threshold $T = 0.55$. The closest possible $\boldsymbol{\beta}$ to $\mathbf{n}$ is $\left(\frac{3}{8}, \frac{3}{8}, \frac{1}{8}, \frac{1}{8}\right)$ corresponding to minimal winning coalitions $(ABC)(ABD)$. However, we claim that this set of minimal winning coalitions is impossible for $T < \frac{2}{3}$. Since (AB) is not winning, we have $w_{AB} := w_A + w_B \leq T$. Since (ACD) is not winning,

$$T \geq w_{ACD} := w_A + w_C + w_D \geq \frac{w_{AB}}{2} + (1 - w_{AB}) \geq 1 - \frac{T}{2} \implies T \geq \frac{2}{3}.$$

Minimum winning coalitions	Banzhaf vector
(A)	$(1,0,0,0)$
(AB)	$(\frac{1}{2},\frac{1}{2},0,0)$
$(AB)(AC)$	$(\frac{3}{5},\frac{1}{5},\frac{1}{5},0)$
$(AB)(AC)(AD)$	$(\frac{7}{10},\frac{1}{10},\frac{1}{10},\frac{1}{10})$
$(AB)(ACD)$	$(\frac{1}{2},\frac{3}{10},\frac{1}{10},\frac{1}{10})$
$(AB)(AC)(BC)$	$(\frac{1}{3},\frac{1}{3},\frac{1}{3},0)$
$(AB)(AC)(BCD)$	$(\frac{5}{12},\frac{1}{4},\frac{1}{4},\frac{1}{12})$
$(AB)(ACD)(BCD)$	$(\frac{1}{3},\frac{1}{3},\frac{1}{6},\frac{1}{6})$
(ABC)	$(\frac{1}{3},\frac{1}{3},\frac{1}{3},0)$
$(ABC)(ABD)$	$(\frac{3}{8},\frac{3}{8},\frac{1}{8},\frac{1}{8})$
$(AB)(AC)(AD)(BCD)$	$(\frac{1}{2},\frac{1}{6},\frac{1}{6},\frac{1}{6})$
$(ABC)(ABD)(ACD)$	$(\frac{2}{5},\frac{1}{5},\frac{1}{5},\frac{1}{5})$
$(ABC)(ABD)(ACD)(BCD)$	$(\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4})$
$(ABCD)$	$(\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4})$

The next closest $\boldsymbol{\beta}$ is $\left(\frac{1}{2},\frac{3}{10},\frac{1}{10},\frac{1}{10}\right)$, which is achievable using $\boldsymbol{w}=\left(\frac{5}{11},\frac{4}{11},\frac{1}{11},\frac{1}{11}\right)$ for $T=0.55$, hence this is the optimal $\boldsymbol{\beta}$. No $\boldsymbol{w}$ with $w_A=w_B$ can have minimum winning coalitions $(AB)(ACD)$ to achieve this $\boldsymbol{\beta}$ since this set of winning coalitions is not symmetric in A,B: it has (ACD) but not (BCD).

Problem 4.

(a) Show that not every 4-candidate preference profile has a (Euclidean) planar embedding.

(b) Show that every finite preference profile admits a metric embedding.

Solution 4. *(a) Assume that there is a planar embedding of the 4 candidates. Consider the complete graph with vertices at the 4 candidates and draw the $\binom{4}{2}=6$ perpendicular bisectors of the edges. Within a region bounded by these 6 bisectors, the preference profile cannot change because any change swaps the ranking of some two candidates and hence crosses a bisector. Since 6 lines divide a plane into at most 22 regions[1], not all $4!=24$ preference profiles can be realized.*

(b) One elegant solution is to let the standard basis vectors in $\mathbb{R}^m$ be the candidates, and consider the standard simplex that they span. Suppose I prefer the first candidate to the second to the third and so on; then I can embed myself in the position $(m,m-1,m-2,\ldots,1)$, normalized to add up to 1 so that it lies on the simplex spanned by the basis vectors. (To do that, scale down by $\binom{m+1}{2}$.) This ensures that I'm closest to my favorite candidate, who sits at $(1,0,0,\ldots,0)$. For any other ranking, permute the coordinates appropriately. This works whether the distances are L^2 or L^1, and it's basically leveraging the Borda embedding.

Similarly, you can leverage the head-to-head embedding in $\mathbb{R}^{\binom{m}{2}}$. This time, every voter is identified with vector of ±1 entries that records their pairwise comparisons, and every candidate is identified with the vector in which they are ranked over the other candidates, and the others are tied (the bullet vote!). For instance, A maps to $(1,1,0)$ in a three-candidate election, while B maps to $(-1,0,1)$ and C maps to $(0,-1,-1)$. It's natural to use L^1 distance, though L^2 will also work. It's easy to see that an $A>B>C$ voter has L^1 distance 1 from A, 3 from B, and 5 from C, which respects their ranking. By symmetry, this works for all rankings.

I'll provide a longer solution to Problem 5 (DW-NOMINATE) separately!

[1] See `https://proofwiki.org/wiki/Number_of_Regions_in_Plane_Defined_by_Given_Number_of_Lines`.